



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Academic Year 2021 – 2022 (Odd Semester)

Details of Programme Organized

Sl. No.	Contents	Details
01.	Title of the programme	Using of Gleeble 3800 Simulator to Shape the Structure and Properties of the Various Engineering Materials and Various Techniques
02.	Programme Theme	Research
03.	Programme Type	Webinar
04.	Details of resource person	Dr. Wojciech Borek Assistant Professor Department of Mechanical Engineering Silesian University of Technology Gliwice, Poland
05.	Mode & Venue / Platform	Google Meet & YouTube
06.	Date & Duration	20.07.2021 & 1 hour
07.	No. of Student Participants	38
08.	No. of Faculty Participants	23
09.	No. of External Participants	15
10.	Expenditure amount (if any)	NIL
11.	Social Media URL	https://youtu.be/VuSG_Zq6iMU
12.	Video URL (if any)	https://youtu.be/VuSG_Zq6iMU

13. Objective of the Programme:

- To impart knowledge on cutting edge technologies (material science) to the participants and to upgrade the expertise of the faculty members in mechanical engineering field.

14. Detail description of the activity:

Extramural Lecture Series on the topic **"Using of Gleeble 3800 Simulator to Shape the Structure and Properties of the Various Engineering Materials and Various Techniques"** was organized by Department of Mechanical Engineering, Ramco Institute of Technology, Rajapalayam on 20.07.2021. The main objective of these ELS is to impart knowledge on research trends in Material Sciences. The Resource Person was Dr.Wojciech Borek, Assistant Professor, Department of Mechanical Engineering, Silesian University of Technology, Gliwice, Poland.

During the session he initially explained the facilities available in Silesian University of Technology at laboratories of Department of Engineering Materials and Biomaterials. Later he specified the facilities available in Gleeble 3800 Simulator. The process simulation capabilities and the material testing applications were explained by the resource person. He explained the complete process with a case example along with photographic results which included simulation result of continuous casting, results of liner thermal expansion using dilatometer, technological plasticity. Further he explained the concept of Powder Metallurgy. Further his knowledge sharing extended to Quasi – HIP, pulsed electric current sintering, spark plasma sintering method (SPS), non- conductive powder processing, etc. At the end, results obtained were presented which gave a much better picture for understating the analysis of various parameters.

The total no. of participants was 76 including internal and external participants. The entire sessions were interactive where faculties raised their doubts and a precise explanation was given by resource person. The overall feedback of program was good. Finally, participants were distributed with e-certificates.

15. Banner / flyer of the programme:



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North vengannallur Village, Rajapalayam - 626 117

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Department of Mechanical Engineering

EXTRAMURAL LECTURE SERIES(ELS) - INT002

**USING OF GLEEBLE 3800 SIMULATOR TO SHAPE THE STRUCTURE
AND PROPERTIES OF THE VARIOUS ENGINEERING MATERIALS AND
VARIOUS TECHNIQUES**



Dr. Wojciech Borek

Assistant Professor

Department of Mechanical Engineering
Silesian University of Technology
Gliwice, Poland

Date : July 20, 2021
(Tuesday)

IST: 01.30 P.M.

Poland Time: 10.00 A.M.

Research Credentials



https://youtu.be/VuSg_Zq6iMU

Registration Link : <https://tinyurl.com/Gleeble3800>

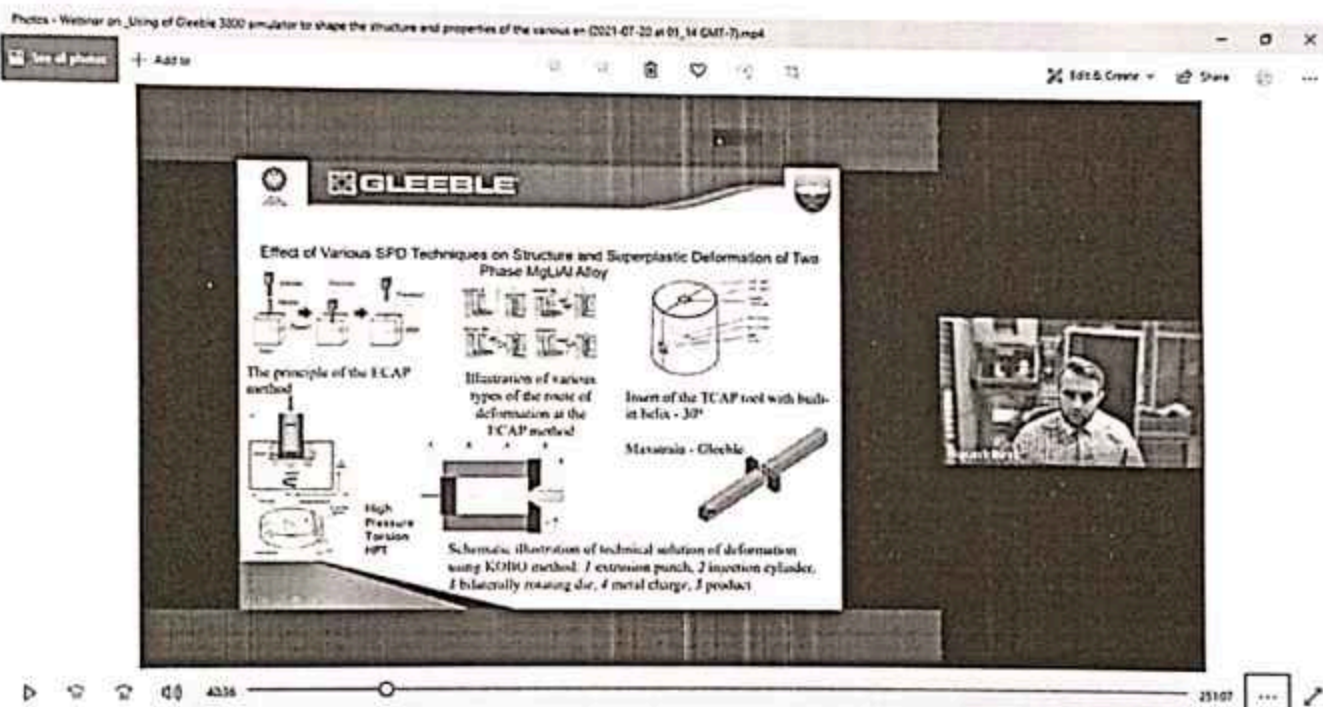
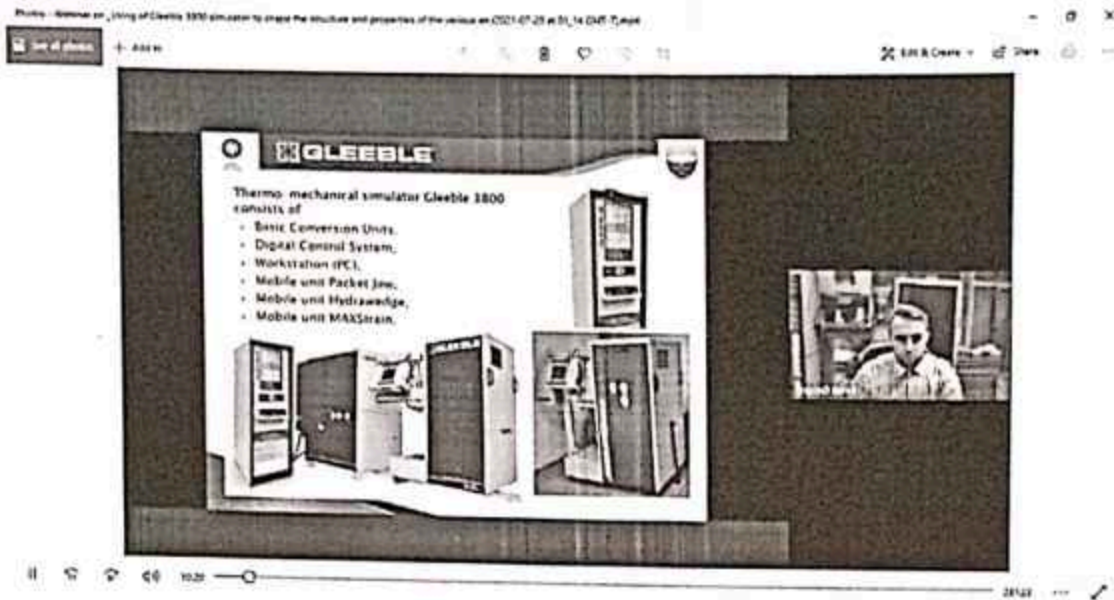
COORDINATOR
Dr.V.SIVAKUMAR
ASSOC. PROF./MECH

CONVENER
Dr.P.SURESHKUMAR
ASSOC. PROF./MECH

PATRON
Dr.S.RAJAKARUNAKARAN
VICE PRINCIPAL & HEAD /MECH

CHIEF PATRON
Dr.L.GANESAN
PRINCIPAL

16. Programme Photographs: (Two Photographs)



Prepared by
26/7/21

Approved by
V. Srinivasan
26/7/21



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